Multilevel Attention Dynamic-Scale Network for HSI and LiDAR Data Fusion Classification

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***Abstract*— Land use/land cover classification with multimodal data has attracted increasing attention. For hyperspectral images (HSIs) and light detection and ranging (LiDAR) data, the combination of them can make the classification more accurate and robust. However, how to effectively utilize their respective strengths and integrate them with the classification task is still a challenging problem. In this article, a multilevel attention dynamic-scale network (MADNet) is proposed. First, in the feature extraction stage, the two modalities are divided into two branches with different scales, which are then fed into the convolutional neural networks (CNNs) to learn shallow features. Then, considering the characteristics of the HSI, a spectral angle attention module (SAAM) with low-level attention is designed to highlight surrounding pixels that have similar spectra to the central pixel of the patch. After that, a dynamic-scale selection module (DSSM) is proposed to screen an appropriate scale for the patches by pixel similarity analysis. Next, combining the Transformer and the CNN, a global–local cross-attention module (GLCAM) is devised to investigate the fused deep-level multimodal features. Distinct from the vanilla Transformer, the GLCAM deploys a distance-weight operator to decrease the redundancies at long distances and effectively reduce misclassifications. Extensive experiments on three paired HSI and LiDAR datasets demonstrate that the proposed MADNet has certain advantages over the existing methods.**

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I. Introduction

With the improvement of remote sensing technology, the modalities of remote sensing data show diversity, such as hyperspectral (HS), multispectral (MS), light detection and ranging (LiDAR), and synthetic aperture radar (SAR). In land use/land cover classification tasks, multimodal data fusion classification methods often yield better results than single-source classification methods [1]. In the existing datasets, HS images (HSIs) and LiDAR are a common multimodal data combination. Specifically, HSI can provide rich spectrum information [2], [3], [4], [5], [6], but it has inevitable spectral issues [7], [8], [9], [10], [11]. Namely, the same species may have different spectral curves due to environmental factors, pests, and radioactive substances. By contrast, two different land covers may exhibit similar spectral characteristics in particular wavelength ranges. Relying solely on the HSI for land cover classification can lead to misclassifications. Fortunately, the elevation characteristics of the LiDAR can partially alleviate the problems [12]. Therefore, research on the HSI and LiDAR data fusion classification has become a hot topic [13], [14].

In recent years, deep learning has been introduced into multimodal data land use/land cover classification tasks due to its powerful learning capability [15], [16], [17], [18], [19]. Convolutional neural network (CNN) is one of the mainstream methods in deep learning. For instance, Hong et al. [20] proposed a general multimodal deep learning framework to provide a baseline solution for the task. Xu et al. [21] designed a two-branch CNN (TBCNN) that extracts spatial–spectral features from HSI and spatial features from LiDAR. Distinct from above single-scale CNN, some studies focus on multiscale features with different objects [22], [23]. Although these studies can capture the features of different scales, the scales are relatively fixed and the computational complexity is high.

With the development of CNN-based network, several studies have added attention mechanism to the network to further improve network performance. For example, squeeze-and-excitation (SE) block and efficient channel attention (ECA) [24] were used in some networks. However, most studies employed universal attention mechanisms to extract

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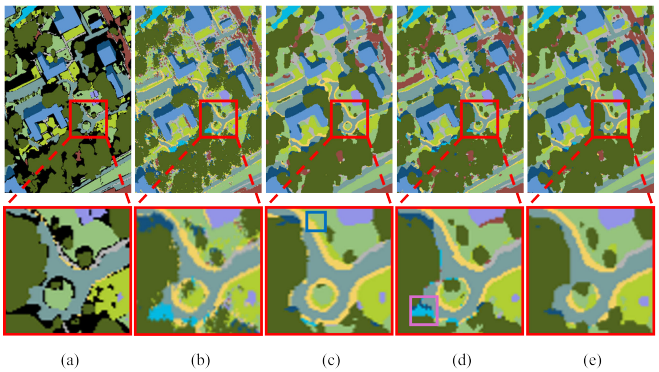


Fig. 1. (a) Ground truth of the MUUFL dataset, and classification maps obtained by (b) TBCNN [21], (c) MFT [25], (d) GLT-Net [29], and (d) our proposed MADNet, respectively.

information from HSI. These universal attention mechanisms are not specifically designed for HSI, leading to an inability to fully utilize spectral information. Moreover, above methods used single-level attention, which was inadequate in extracting multiple layers of rich features required for multimodal classification.

Except for the CNN, another fundamental deep learning model, vision transformer (ViT), has been introduced due to its powerful long-range dependency capability. For example, Roy et al. [25] proposed a multimodal fusion Transformer (MFT) network to fusion the HSI and LiDAR data. Moreover, with the development of the two basic models, it has been verified that integrating CNN and Transformer together can further improve network performance [26], [27], [28]. Ding et al. [29] designed a global–local Transformer network (GLT-Net) to capture complex spatial–spectral information with the CNN and Transformer.

To illustrate the differences between CNN, Transformer and their combinations, Fig. 1(a)–(e) shows the local magnification of the ground truth, classification maps of the CNN-based TBCNN [21], Transformer-based MFT [25], hybrid CNN and Transformer-based GLT-Net [29], and the proposed multilevel attention dynamic-scale network (MADNet), respectively. Because CNN focuses more on capturing local features and less on global information, the classification result of TBCNN is not that good. While the methods of employing a vanilla Transformer and integrating the Transformer with CNN have been shown to be more effective than using CNN alone in this experiment, they still encounter misclassification problems, as evidenced by the blue and pink boxes in Fig. 1.

In summary, although the aforementioned studies have made some progress in multimodal classification, there are still limitations in existing research, which include the following.

1) In the field of HSI and LiDAR fusion classification, most attention-related studies tend to focus on using channel attention and spatial attention during shallow feature extraction processes. Although channel attention demonstrates broad applicability across diverse modalities, its effectiveness in HSI appears constrained. Furthermore, many methods have not highly focus on the impact of incorporating attention at various levels within the network.
2) Land use/land cover classification typically involves objects of various sizes. Previous networks predominantly utilized single-scale or fixed multiscale patches as inputs. However, single-scale patches struggle to accommodate size variations across different object classes, while fixed multiscale may unnecessarily increase the network complexity.
3) The Transformer has remote dependency capability to improve the classification performance. However, recent studies [30], [31] and Fig. 1(d) indicate that the vanilla Transformer introduces remote redundant information, leading to sudden misclassified changes in continuous classification results.

To address the above issues, we propose MADNet for the fusion classification of HSI and LiDAR data. Within this framework, the MADNet leverages attention-based modules to explore the features of HSI and LiDAR data at multiple levels. Initially, a dual-branch network for HSI-LiDAR is used for feature extraction, with the HSI branch emphasizing low-level spectral information through spectral angle attention modules (SAAMs). Subsequently, a dynamic-scale selection module (DSSM) is introduced to choose appropriate patch scales by analyzing the pixel similarity of HSI. To maintain resolution consistency, the patch size of LiDAR matches that of HSI. The two scales are then fused in the global–local cross-attention module (GLCAM) to further focus on high-level features. Finally, the classification module outputs the classification results.

The main contributions of this study are as follows.

1) We propose a multilevel attention mechanism to extract effective information from two modalities. For the low level, the SAAM is designed to utilize the spectral relationship between the central pixel and its neighbors, which is more effective on HSI than universal attention. For the high level, the cross-attention in the GLCAM focuses on deep features after modality fusion. And our designed multilevel attention can play a coordinated role in different depth features.
2) To better adapt to objects of different sizes in classification tasks, we design the DSSM. This module can adaptively choose patches of different scales based on the pixel similarities. Specifically, if the pixel similarity reaches the threshold, it proves that the small-scale patches are suitable with less computing loss. Conversely, the large-scale patches should be considered because continuous pixels with the same class as the central pixel may exist in large-scale scenes.
3) In GLCAM, we propose a Transformer with a special operator called distance-weight-based Transformer (DWBT). This operator complements the relationships between neighboring pixels while reducing the interference of distant redundant information.
4) The proposed MADNet is evaluated using three benchmark remote sensing datasets, namely Houston2013, Trento, and MUUFL. The experimental results consistently demonstrate the superiority of the method.

The rest of the article is structured as follows. Section II summarizes the existing works on attention mechanism,

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multiscale feature extraction, and ViT. Section III details our proposed network. Section IV introduces the datasets, parameter setting, ablation, and comparison experiments. Section V gives the conclusion.

II. Related Work

A. Attention Mechanism

In computer vision, the attention mechanism mimics human visual and cognitive systems [32], which focuses on specific regions rather than the entire image. By incorporating the mechanism, deep learning networks can automatically attend to important information in the input, thereby improving the model performance and generalization capabilities [33].

Attention mechanism was originally applied in the general computer vision framework with various downstream tasks. For instance, Hu et al. [34] proposed an SE network (SENet) that adaptively attends to channel-wise features. Although this strategy is widely used in classification tasks, dimensionality reduction has a negative effect on the attention prediction. Woo et al. [35] proposed a convolutional block attention module (CBAM), which obtains attention maps along both the spatial and channel dimensions. This method conquers the side effects caused by the dimensionality reduction to some degree, but the calculation cost of this network is high. To solve this issue, Wang et al. [36] proposed an ECA module that reduces the model complexity while maintaining the performance through a non-dimensional reduction local cross-channel interaction strategy. Gao et al. [37] proposed an enhanced unmixing-inspired unsupervised network with attention-embedded degradation learning to realize MS-aided HS super-resolution. Additionally, Hou et al. [38] introduced a coordinated attention mechanism that decomposes channel attention into one-dimensional feature encoding along two spatial directions, preserving positional information while capturing long-range dependencies. However, the network focuses too much on coordinate information and is not flexible enough for multimodal classification tasks.

With the development of multimodal remote sensing, attention mechanism is gradually applied to feature extraction of different modalities. Dong et al. [39] introduced a self-guided cross-attention multibranch feature fusion network, which utilizes attention within and between two modalities for the feature fusion. Zhang et al. [40] proposed a multilevel attention fusion network that highlights useful spatial information in each modality through a mutual guidance. Shi et al. [41] introduced a threshold gated-cross aggregation network that incorporates feature fusion through cross-attention modules. Feng et al. [42] proposed an HSI-LiDAR classification method based on multiview feature learning and multilevel information fusion, which effectively enhances the classification performance. Xiu et al. [24] utilize ECA to extract the shallow features of two modalities as the input of the graph neural networks. Yang et al. [43] proposed a LiDAR-guided cross-attention fusion. The method used LiDAR data as a "query" to search and identify "keywords" from the HSI to select the relevant bands.

Although the above attention-based methods have achieved promising improvements, most of them do not perform well in the case of complex spectra due to a lack of consideration of the high-dimensional natural characteristics of the HSI [44]. Moreover, few studies have focused on whether adding different attention to various levels of the network and forming multilevel attention can further improve network performance.

B. Multiscale Feature Extraction

In land use/land cover classification tasks, feature extraction is the fundamental function of a network, and how to extract valid features is a key issue. Specifically, single-scale feature extractor often leads to redundancy or information loss due to the inappropriate scale [45]. Multiscale feature extraction can address the issue by exploring the features at different spatial and spectral scales [46]. Particularly, multiscale feature extraction methods decompose the spectral, textural, shape, and structural information into different frequency bands, scales, and resolutions to obtain features at various views, which improves the classification accuracy and robustness. For instance, in [22], a multiscale fusion module was designed for extract richer spectral information, and the adaptive feature fusion module was designed for the fusion of features from diverse sources in multisource data fusion. In [23], a pyramidal multiscale convolutional network was designed to extract spatial geometric structure features. Wang et al. [47] proposed a dual-branch CNN that integrates shallow and deeper convolutional features during spatial feature extraction, allowing the network to focus more on global information. Wang et al. [48] proposed a multiscale feature interactive model, which obtains richer scale information with limited parameters, achieving cross-fusion of multimodal data from both local and global perspectives.

However, for different scales of various ground objects, single-scale patches are often difficult to adapt to size changes. Moreover, the multiscale network greatly increases the amount of computation. Therefore, we propose the DSSM to dynamically select scales for objects.

C. Vision Transformer

Transformer was introduced by Vaswani et al. [49] and is primarily designed for natural language processing (NLP). The core idea behind Transformers is the self-attention mechanism, which captures long-range dependencies in the data and enables them to excel in sequential data processing. Dosovitskiy et al. [50] adapted the Transformer to computer vision and proposed the ViT, which achieves comparable and even better performance than the dominant CNNs on the image classification tasks.

Like Transformers in NLP, ViT utilizes the self-attention mechanism to capture global contextual information in images, which allows the model to handle long-range dependencies within the image and learn interactions between different regions [51], [52], [53], [54]. Specifically, the image is first divided into non-overlapping patches, which are then linearly embedded into high-dimensional vectors. Then, positional encoding is imposed on the patch embedding to involve

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the order of the patches. After that, a Transformer encoder processes the resulting sequence to capture local and global contextual information, improving remote dependency of the pixels and enhancing the classification performance.

Due to the remote dependence of Transformer, many researchers use Transformer alone or in combination with CNN to increase the performance of remote sensing classification tasks. For example, Gao et al. [55] presented a Transformer-based spatial–spectral mixer to achieve cross-scale calibration with original-scale images. Yu et al. [56] developed a cross-context capsule ViT for HSI and LiDAR classification, which can effectively fuse cross-context features. Zhang et al. [57] designed different Transformers to extract pixelwise feature embedding relationships in a global manner and shared spatial features for the two modalities, respectively. Zhao et al. [58] proposed a dual-branch network combining them, which employs a cross-token attention fusion encoder to integrate HSI-LiDAR heterogeneous information. Tu et al. [59] used a CNN-based structure to obtain high-frequency features and a Transformer-based block for low-frequency features.

However, recent studies clarify that ViT introduces redundant long-distance information, which produces uneven misclassifications in the classification results. Therefore, we improve the vanilla Transformer by adding a DWBT to reduce the side effects of the remote redundancies.

III. Methodology

A. Overall Architecture

The MADNet consists of several convolutional modules (CMs), SAAM, DSSM, GLCAM, and the classification module, which is illustrated in Fig. 2. First, the shallow features of two different scale patches from the HSI and LiDAR are extracted by the CMs. Subsequently, the SAAM highlights the significant low-level feature of the HSI via increasing the weight of surrounding pixels with similar spectra to the central pixel. Next, the DSSM can adaptively select the patch of the HSI with an appropriate scale by analyzing the similarity statistics. For resolution consistency, the scale of the LiDAR is consistent with that of the HSI. After the DSSM is another CM for further feature extraction, and then the two modalities are integrated by the concatenate operation. After that, the GLCAM is designed to mine the fused multimodal high-level information. In specific, the proposed DWBT in the GLCAM reduces the remote redundant information and improves the classification efficiency. Finally, the comprehensive features are sent into the classification module to obtain the final classification results. Notably, the SAAM and the GLCAM form the multilevel attention in the framework, which can effectively utilize various dimensions of the information from the HSI and LiDAR data. In the following, we will elaborate on the portions of the frameworks in detail.

B. Convolutional Module

The network has three levels of CMs: the first level is at the initial feature extraction, the second level is located after the DSSM, and the last is in the GLCAM. The principle of the first-level CM is as follows. Let us denote the HSI and LiDAR features as $\mathbf{X}_{\mathrm{H}} \in \mathbb{R}^{H\times W\times C_{\mathrm{H}}}$ and $\mathbf{X}_{\mathrm{L}} \in \mathbb{R}^{H\times W\times C_{\mathrm{L}}}$, respectively, where H and W represent the spatial dimensions and C_{H} and C_{L} represents the number of channels in the multimodal data. To capture the multiscale spatial information, two scales of patches denoted as $\mathbf{X}_{\mathrm{H1}} \in \mathbb{R}^{\mathrm{ps}_1\times \mathrm{ps}_1\times C_{\mathrm{H}}}$ and $\mathbf{X}_{\mathrm{H2}} \in \mathbb{R}^{\mathrm{ps}_2\times \mathrm{ps}_2\times C_{\mathrm{H}}}$ are extracted from the HSI data. Correspondingly, $\mathbf{X}_{\mathrm{L1}} \in \mathbb{R}^{\mathrm{ps}_1\times \mathrm{ps}_1\times C_{\mathrm{L}}}$ and $\mathbf{X}_{\mathrm{L2}} \in \mathbb{R}^{\mathrm{ps}_2\times \mathrm{ps}_2\times C_{\mathrm{L}}}$ at two scales are extracted from the LiDAR data. The four patches are then fed into the CMs, which are used to extract shallow features from the images. Each CM comprises two convolutional blocks, and each block includes a 2-D convolution (Conv2d) with a kernel size of 3×3, batch normalization (BN), and rectified linear unit (ReLU) activation. The number of convolutional kernels is empirically set to 64 for both blocks. The ReLU activation function is chosen due to its sparsity property, effectively addressing the vanishing gradient problem and accelerating the network training. The outputs of the CMs are four patches from the two modalities, denoted as $\mathbf{X}'_{\mathrm{H1}} \in \mathbb{R}^{\mathrm{ps}_1\times \mathrm{ps}_1\times 64}$, $\mathbf{X}'_{\mathrm{H2}} \in \mathbb{R}^{\mathrm{ps}_2\times \mathrm{ps}_2\times 64}$, $\mathbf{X}'_{\mathrm{L1}} \in \mathbb{R}^{\mathrm{ps}_1\times \mathrm{ps}_1\times 64}$ and $\mathbf{X}'_{\mathrm{L2}} \in \mathbb{R}^{\mathrm{ps}_2\times \mathrm{ps}_2\times 64}$. The principles of the other two levels of CM are same as the first level, with variations in the channels.

C. Spectral Angle Attention Module

Although various attention mechanisms have been developed and proven effective in remote sensing, little effort has been devoted to exploring the spectral relationships between the central and the surrounding pixels of the patch. In [60], contrast learning is used to reduce the spatial distance of neighbor pixels. Inspired by the article, we design the SAAM to explore the spectral distance of the neighbor pixel from the central pixel and translate this distance into attention weights. And we use the spectral angle between pixels to measure their spectral distance.

The module block diagram is depicted in Fig. 3, where the pixels with darker colors indicate higher spectral similarity with the central pixel. Specifically, the spectral angle between each pixel on the patch and the central pixel is calculated by the following formula:

$$\begin{aligned} s_{ij} &= \mathrm{SAM}(\mathbf{x}_{ij}, \mathbf{x}_{\mathrm{center}}) \\ &= \arccos\left(\frac{\mathbf{x}_{ij}^T\mathbf{x}_{\mathrm{center}}}{\sqrt{\mathbf{x}_{ij}^T\mathbf{x}_{ij}}\sqrt{\mathbf{x}_{\mathrm{center}}^T\mathbf{x}_{\mathrm{center}}}}\right) \end{aligned} \tag{1}$$

where $\mathbf{x}_{ij}$ represents the pixel in the ith row and jth column of $\mathbf{X}'_{\mathrm{H1}}$ or $\mathbf{X}'_{\mathrm{H2}}$ with 64 channels. While $\mathbf{x}_{\mathrm{center}}$ represents the central pixel of $\mathbf{X}'_{\mathrm{H1}}$ or $\mathbf{X}'_{\mathrm{H2}}$. T represents the transpose operation. $\mathbf{s}_{ij}$ represents the spectral angle between the $\mathbf{x}_{ij}$ and $\mathbf{x}_{\mathrm{center}}$. Then, we have the following output weight of each pixel:

$$w_{ij} = e^{-\mathrm{sigmoid}(\gamma s_{ij})} \tag{2}$$

where γ is a coefficient that can be learned by the network, with an initial value of 0. The role of the sigmoid$(\cdot)$ function is to map the output of the previous layer to a range between

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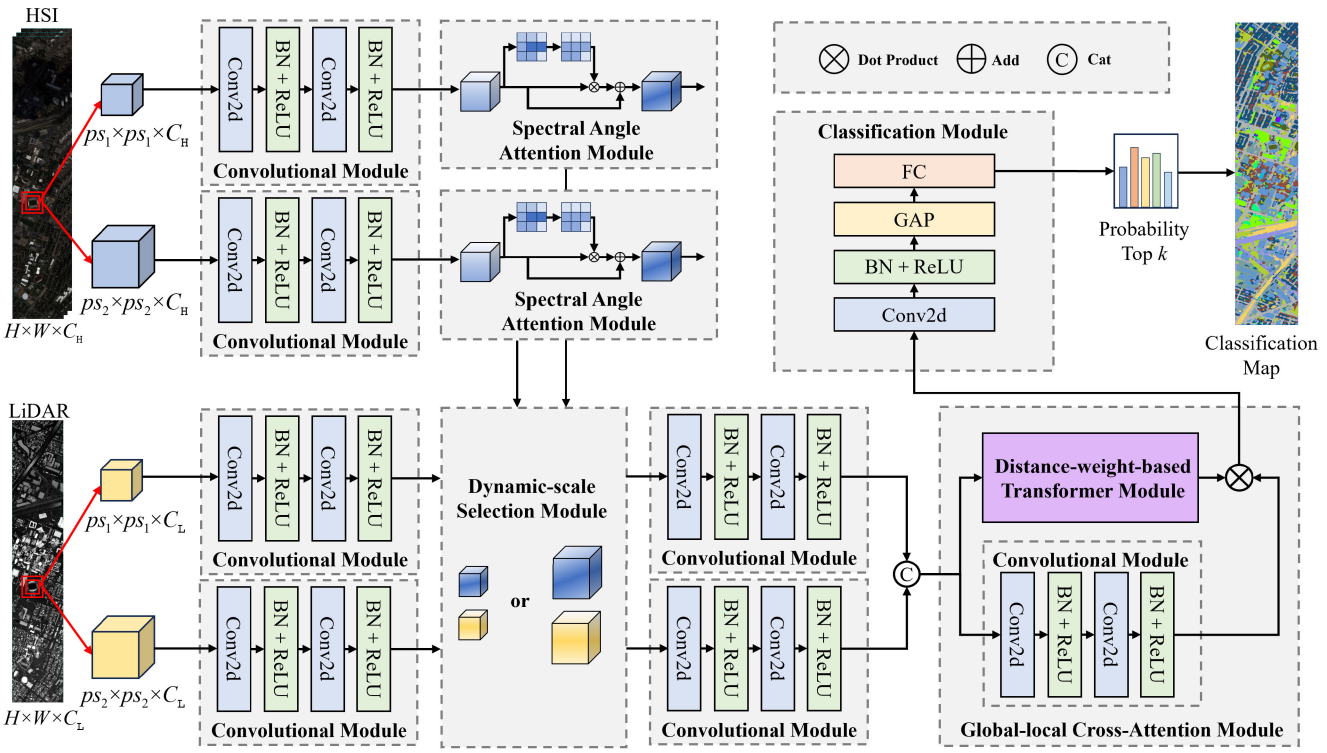


Fig. 2. Flowchart of the proposed MADNet, which comprises several CMs, SAAM, DSSM, GLCAM, and the classification module.

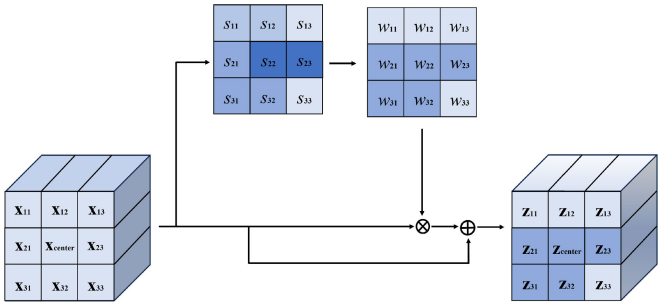


Fig. 3. Illustration of the proposed SAAM.

0 and 1, which effectively normalizes the values and prevents extreme values from affecting the computations significantly. The $e^{-(\cdot)}$ function amplifies the weights of small spectral angles. w_{ij} represents the weight of pixel $\mathbf{x}_{ij}$.

After that, the weight result is multiplied with the input, and a residual structure is introduced to get the final output. The processing procedure can be expressed as

$$\mathbf{z}_{ij} = \mathbf{x}_{ij} w_{ij} + \mathbf{x}_{ij}. \tag{3}$$

The output of the SAAM consists of feature cubes $\mathbf{Z}_{\mathrm{H1}}$ and $\mathbf{Z}_{\mathrm{H2}}$, corresponding to the small-scale and large-scale patches, respectively. $\mathbf{z}_{ij}$ refers to the pixel in the $\mathbf{Z}_{\mathrm{H1}}$ and $\mathbf{Z}_{\mathrm{H2}}$.

D. Dynamic-Scale Selection Module

For the scale diversity of different objects, the DSSM determines the appropriate scale by calculating the pixel similarities within the patch of the HSI features. The size of the LiDAR features is determined to be consistent with that of the HSI.

First, the average pixel of small-scale HSI patch is calculated as

$$\mathbf{T}^c = \mathrm{avg}\left(\mathbf{Z}^c_{\mathrm{H1}}\right) \tag{4}$$

where c represents the cth channel of $\mathbf{Z}_{\mathrm{H1}}$. Then, the difference between the average and the input value can be computed by

$$\mathbf{Y}^c = \left|\mathbf{Z}^c_{\mathrm{H1}} - \mathbf{T}^c\right| \tag{5}$$

where $|\cdot|$ represents the absolute value. If the elements in $\mathbf{Y}^c$ are less than a predefined parameter α, a counter is incremented. It can be represented as

$$\mathrm{counter} = \begin{cases} \mathrm{counter} + 1, & y^c_{ij} \le \alpha \\ \mathrm{counter}, & y^c_{ij} > \alpha \end{cases} \tag{6}$$

where y^c_{ij} is the pixels of $\mathbf{Y}^c$ and the initial value of the counter is 0.

If the counter exceeds the threshold β, it indicates that there is high similarity between the pixels at this scale, and it should continue to be used. Otherwise, if the pixel similarity within the current patch size is low, it implies that a cluttered scene with multiple object categories makes classification difficult. Considering that larger patches may contain more pixels of the same category as the central pixel, it is advisable to choose larger patches. The size of LiDAR patches should be consistent with HSI to prevent interference from multimodal fusion caused by different resolutions. And then the patches of the two modalities are simultaneously input into the next CM. The outputs of the CM are denoted as $\mathbf{F}_{\mathrm{H}} \in \mathbb{R}^{\mathrm{ps}\times\mathrm{ps}\times C_{\mathrm{F}}}$

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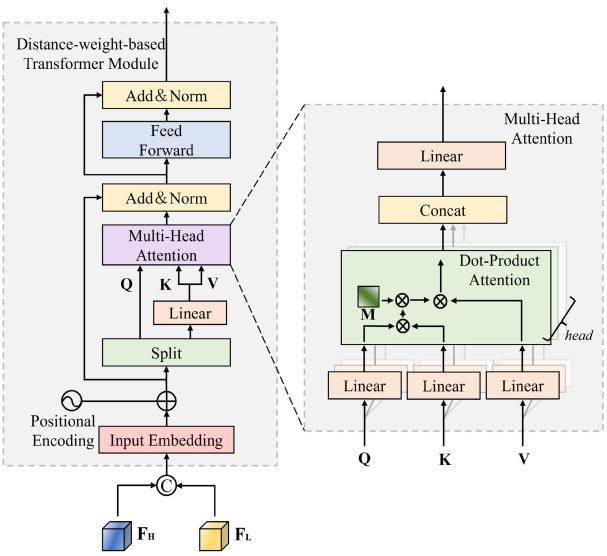


Fig. 4. Left part of the figure provides a detailed description of the DWBT, while the right part shows the internal structure of the multihead attention module.

and $\mathbf{F}_{\mathrm{L}} \in \mathbb{R}^{\mathrm{ps}\times\mathrm{ps}\times C_{\mathrm{F}}}$. Here, ps represents the patch size after scale selection, and C_{F} represents the number of channels.

E. Global–Local Cross-Attention Module

To further investigate the deep-level multimodal features, the GLCAM comprises a CM and an innovative DWBT, which combines the local feature extraction ability of the CNN and the long-range dependency capability of the Transformer. Notably, the Transformer branch does not directly output the classification results but provide weight relationships between pixels, which are multiplied with the CNN outputs.

As shown in Fig. 4, before the Transformer branch, the features $\mathbf{F}_{\mathrm{H}}$ and $\mathbf{F}_{\mathrm{L}}$ are fused by concatenation. Then the fused features are embedded by the input embedding block. After that, positional encoding is added to the features and the features are split into two branches. One corresponds to the HSI data, denoted as $\mathbf{Q} \in \mathbb{R}^{\mathrm{ps}^2\times C_{\mathrm{E}}}$, the other forms $\mathbf{K} \in \mathbb{R}^{\mathrm{ps}^2\times C_{\mathrm{E}}}$ and $\mathbf{V} \in \mathbb{R}^{\mathrm{ps}^2\times C_{\mathrm{E}}}$ through the linear block, which are related to the LiDAR data. Here, C_{E} is the number of channels in this stage, which is set to 64 in the article.

Next, the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are fed into the multihead attention module for further fusion, which is detailed at the right of Fig. 4. In particular, the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are mapped through a linear module to obtain $\mathbf{Q}_{\mathrm{h}}$, $\mathbf{K}_{\mathrm{h}}$, and $\mathbf{V}_{\mathrm{h}}$, respectively, where h represents the hth head. The cross product of $\mathbf{Q}_{\mathrm{h}}$ and $\mathbf{K}_{\mathrm{h}}$ generates the relationship map $\mathbf{A}_{\mathrm{h}}$ between pixels in the hth head. Noticeably, considering that the distance between pixels affects their interactions, we introduce the distance weights that are calculated as

$$\mathbf{M}_{\mathrm{h}}^{i,j,m,n} = \lambda_{\mathrm{h}} e^{-\frac{(i-m)^2+(j-n)^2}{2\theta^2\mathrm{ps}^2}} \tag{7}$$

where i and j represent the coordinates of a pixel on the patch. m and n represent the coordinates of the interacting pixel on the patch. θ is a learnable parameter. λ_{h} represents the multihead coefficient, which is a parameter learned by the network and initialized to 0.

Although the Transformer can establish long-range dependencies, the increased distance also brings redundant information. The purpose of this step is to learn the relationship between all pixels in the patch and focus more on the information close the central pixel while suppressing long-range redundancies. By multiplying $\mathbf{A}_{\mathrm{h}}$, $\mathbf{M}_{\mathrm{h}}$ and $\mathbf{V}_{\mathrm{h}}$, the weights between pixels in the hth head can be obtained. The entire process can be described as

$$\mathrm{DPA}(\mathbf{Q}_{\mathrm{h}}, \mathbf{K}_{\mathrm{h}}, \mathbf{V}_{\mathrm{h}}) = \mathrm{softmax}\left(\frac{\mathbf{Q}_{\mathrm{h}}\mathbf{K}_{\mathrm{h}}^{T}\mathbf{M}_{\mathrm{h}}}{\sqrt{d/\mathrm{head}}}\right)\mathbf{V}_{\mathrm{h}} \tag{8}$$

where d is a scaling factor set to 64, and head is the number of heads. Then, the multihead attention is obtained by concatenating and linearizing the multihead weight relations.

After normalized and passed through a feed-forward operation, the output of the Transformer is multiplied with the convolutional branch to obtain the output of GLCAM.

F. Classification Module

The classification module selects the class with the highest probability as the final classification result and outputs the corresponding feature map. It consists of the Conv2d, BN, ReLU, global average pooling (GAP), and fully connected (FC) layers. This module can be divided into three blocks. The first block sequentially includes a Conv2d with 1×1 kernel size, BN, and ReLU activation operations. The number of convolutional filters is set to 32 to reduce the number of channels. The second block performs GAP to reduce the spatial dimensions to 1×1 and captures pixel-level category information. The third block is an FC layer that maps the channels to the predefined number of land cover classes Num. By applying the softmax function, the outputs of classification module are transformed into Num-dimensional probabilities ranging from 0 to 1, which is donated as $\mathbf{P}$. Then, the log function is introduced to facilitate calculations, and we use cross-entropy to represent the loss function. The entire process is represented as

$$\mathrm{loss}(\mathbf{P}, y) = -\log\left(\frac{e^{\mathbf{P}_y}}{\sum_{l=1}^{\mathrm{Num}} e^{\mathbf{P}_l}}\right) \tag{9}$$

where y is the label of the current pixel. $\mathbf{P}_y$ represents the predicted probability of the yth category.

IV. Experiments and Analysis

In this section, we conduct experiments on three widely adopted HSI and LiDAR datasets to evaluate the capability of the proposed method and compare it with other advanced models.

A. Data Description

1) Houston2013 Dataset: This dataset was collected by the National Center for Airborne Laser Mapping (NCALM) in June 2012 at the University of Houston campus and the surrounding areas in Texas, USA. The dataset consists of a Digital Surface Model (DSM) derived from the HSI and LiDAR data. The HSI data consists of 349 × 1905 pixels with

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS IN HOUSTON2013 DATASET

No.	Class	Training	Test
1	Health Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS IN TRENTO DATASET

No.	Class	Training	Test
1	Apple Trees	129	3905
2	Building	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052

144 spectral channels. It has a spatial resolution of 2.5 m/pixel and wavelength ranging between 0.38 and 1.05 μm. LiDAR-derived DSM has the same size and spatial resolution as HSI. The IEEE GRSS Data Fusion Competition provides the data and the ground truth with 15 land use/land cover classes. Table I shows the details of the training and test sample counts.

2) Trento Dataset: This dataset was collected in a rural area in the southern part of Trento. It consists of a DSM derived from HSI and LiDAR data. The HSI data was acquired using the AISA Eagle sensor, which consists of 600 $\times$ 166 pixels with 63 spectral channels. The spectral range of the channels is from 0.42 to 0.99 μm. The LiDAR data was collected using the optical ALTM 3100 EA sensor. Both the HSI and LiDAR-derived DSM have a spatial resolution of 1 m. This dataset focuses on land cover classification tasks for six different categories. Table II provides the training and test sample counts for the Trento dataset.

3) MUUFL Dataset: This dataset was captured in November 2010 over the Gulf Park campus of the University of Southern Mississippi. It consists of a DSM derived from HSI and LiDAR data. The HSI data was acquired using the ITRES CASI-1500 sensor with 64 available spectral channels ranging from 0.38 to 1.05 μm and a spatial resolution of 0.54 $\times$ 1.0 m. The LiDAR data was collected using the Gemini Airborne LiDAR System with a spatial resolution of 0.60 $\times$ 0.78 m. This dataset contains a total of 53 687 ground-truth samples for 11 different categories. Table III provides the training and test sample counts for the MUUFL dataset.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS IN MUUFL DATASET

No.	Class	Training	Test
1	Trees	100	23146
2	Mostly Grass	100	4170
3	Mixed Ground Surface	100	6782
4	Dirt and Sand	100	1726
5	Road	100	6587
6	Water	100	366
7	Building Shadow	100	2133
8	Building	100	6140
9	Sidewalk	100	1285
10	Yellow Curb	100	83
11	Cloth Panels	100	169

B. Parameter Tuning

The experimental equipment is a Lenovo computer with an RTX3060 graphics card, 6 GB of video random access memory (VRAM). The version of CUDA is 11.7 and the PyTorch is 2.0.1. We use an Adam optimizer with a learning rate decay parameter of 0.9. We empirically set the batch size to 64 because the same settings were used in most of the comparative experiments and achieved excellent performance. The number of training epochs is set to 500. To objectively reflect the classification performance, we use overall accuracy (OA), average accuracy (AA), and Kappa coefficient as the evaluation criteria. All the evaluation metrics take the average of ten classification results.

1) Multiscale Input Patch Size: The input patch size has significant impacts on the model performance. We use different patch combinations to adapt to changes in object size. This combination includes two scales, small and large, and our network can adaptively select small or large-scale patches from the patch combination. The experimental results are shown in Fig. 5. It can be found that combinations with a margin of 2 perform better than those with a larger margin. This is likely due to the significant discrepancy between the two scales, which results in undesirable interference. As a result, we select two adjacent patch sizes as inputs. As shown in Fig. 5(a), for the Houston2013 dataset, the most suitable small-scale patch size was 11 $\times$ 11. In the case of the Trento and the MUUFL datasets, depicted in Fig. 5(b) and (c), the optimal small-scale patch size is found to be 9 $\times$ 9. The variations in optimal patch sizes are attributed to differences in the resolution and types of objects present in each dataset.

2) Dynamic-Scale Feature Discriminator Parameter: In Section III, it is mentioned that parameter α represents the similarity threshold of pixels on a patch. If the difference between a pixel and the mean pixel is smaller than α, it indicates a high similarity with the surrounding pixels.

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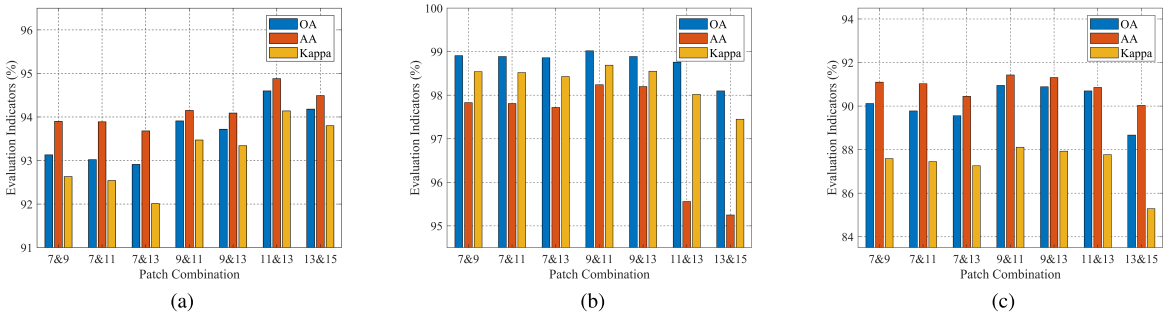


Fig. 5. Comparison of model performance across varying patch sizes. (a) Houston2013. (b) Trento. (c) MUUFL.

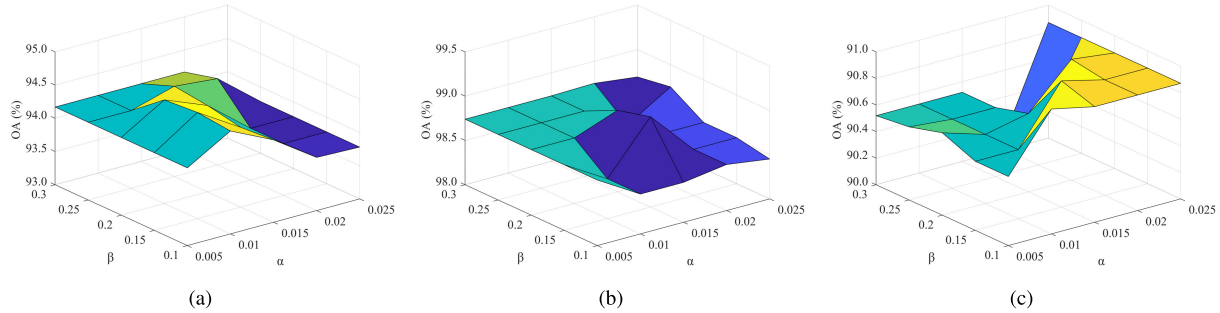


Fig. 6. Comparison of model performance across varying parameters. (a) Houston2013. (b) Trento. (c) MUUFL.

Parameter β represents the statistical threshold for similar pixels. If the number of similar pixels on a patch reaches β, it suggests that the current scale is appropriate. Conversely, if the number of similar pixels on a patch is low, a larger patch scale should be chosen to increase the useful information. We use the grid search strategy to find the appropriate value of α and β [61], [62]. In Fig. 6, it can be observed that the classification accuracy decreases when α and β values are too small or too large. A small α value excludes similar pixels from the statistics, while a large α value includes dissimilar pixels in the statistics. A large β value fixes the scale selection to a larger scale, whereas a small β value fixes it to a smaller scale. According to experimental results, α should be set to 0.01 and β should be set to 0.15 for the Houston2013 dataset. While α should be set to 0.015 and β should be set to 0.15 for the Trento dataset and the MUUFL dataset.

3) Learning Rate: As an important hyperparameter of deep learning networks, the learning rate plays an important role in the optimization procedure. If the learning rate is too small, the convergence process will become very slow. When it is set too large, the gradient may oscillate around the minimum value, and the network may not even converge. Therefore, we try to find the appropriate learning rate through experiments. Fig. 7 shows the classification performance of the framework across the three datasets by different learning rates {0.001, 0.0005, 0.0001, 0.00005}. As a result, the optimal learning rate of the three datasets is 0.0001.

C. Ablation Studies

To verify the validity of the multilevel attention and dynamic scale of the MADNet, we performed ablation experiments on the three datasets. Notably, the multilevel attention ablation experiment involves two modules: the first module is SAAM designed for low-level intrinsic spectrum information, and the second module is DWBT designed for high-level fused multimodal information. While the module involved in the dynamic-scale ablation experiment is the DSSM. Ten times independent experiments were carried out in each group, and the results are expressed as mean and standard deviation.

1) Multilevel Attention Ablation Experiments: In this section, we performed ablation experiments for SAAM and GLCAM, as shown in Table IV. The GLCAM consists of a CM and DWBT. And DWBT comprises a distance weight operator M. In this experiment, we remove and retain the SAAM, CM, DWBT, and M, respectively. The first row in Table IV shows the experiment results of the baseline removing both SAAM and GLCAM. The second row shows the experiment results of the baseline with SAAM. The third row shows the experiment results of the baseline with CM. The fourth row shows the experiment results of the baseline with GLCAM (without the operator M). The fifth row shows the experiment results of the baseline with the complete GLCAM. The last row shows the experiment results of the complete network with both SAAM and GLCAM. It can be seen that using SAAM or GLCAM is better than the baseline, which means that the design of attention at each level plays a significant role. Using both SAAM and GLCAM, the results show that multilevel attention can focus on both low-level natural spectrum features and high-level fusion multimodal information, and they can coordinate each other to make the network performance the best. In addition, the ablation experiments for the operator M show that it can further improve the performance of the network.

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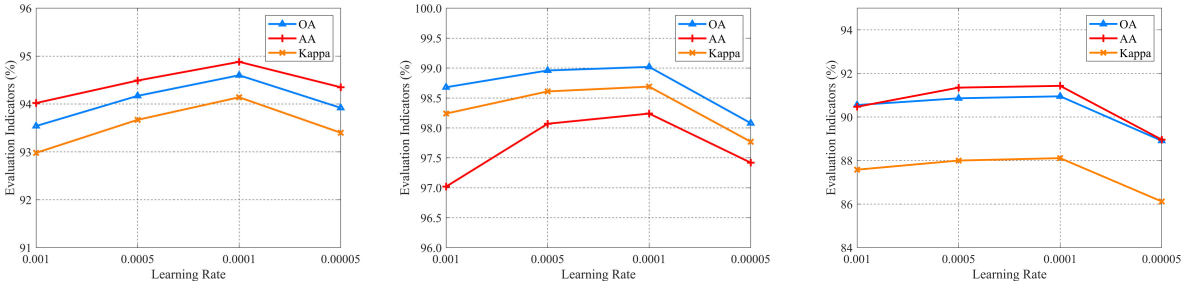


Fig. 7. Comparison of model performance across varying learning rates. (a) Houston2013. (b) Trento. (c) MUUFL.

TABLE IV
ABLATION ANALYSIS OF SAAM AND GLCAM IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON HOUSTON2013, TRENTO, AND MUUFL DATASETS

Module				Houston2013			Trento			MUUFL		
SAAM	CM	DWBT	operator M	OA (%)	AA (%)	Kappa (×100)	OA (%)	AA (%)	Kappa (×100)	OA (%)	AA (%)	Kappa (×100)
×	×	×	×	92.01±00.51	92.42±00.49	91.62±00.54	97.03±00.18	96.58±00.82	96.98±00.29	87.67±00.42	88.16±00.66	85.82±00.59
✓	×	×	×	93.15±00.37	93.38±00.32	92.65±00.38	97.75±00.15	97.46±00.58	97.68±00.29	88.83±00.42	89.26±00.35	86.52±00.53
×	✓	×	×	92.03±00.54	92.50±00.53	91.61±00.58	97.05±00.18	96.64±00.68	97.02±00.28	87.85±00.38	88.21±00.35	85.96±00.47
×	✓	✓	×	92.93±00.57	93.62±00.56	92.03±00.65	97.68±00.17	96.79±00.83	97.19±00.26	88.62±00.39	88.73±00.56	86.21±00.53
×	✓	✓	✓	93.86±00.31	94.35±00.25	93.40±00.28	98.02±00.18	97.42±00.51	98.05±00.28	89.87±00.38	90.36±00.35	87.32±00.47
✓	✓	✓	✓	**94.60±00.33**	**94.88±00.29**	**94.14±00.36**	**99.02±00.12**	**98.24±00.33**	**98.69±00.16**	**90.95±00.30**	**91.43±00.70**	**88.11±00.37**

TABLE V
ABLATION ANALYSIS OF THE DYNAMIC SCALE IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON HOUSTON2013, TRENTO, AND MUUFL DATASETS

Datasets	Houston2013			Trento			MUUFL		
Indicators	OA (%)	AA (%)	Kappa (×100)	OA (%)	AA (%)	Kappa (×100)	OA (%)	AA (%)	Kappa (×100)
Small Patch	93.78±00.31	94.23±00.28	93.34±00.33	98.52±00.14	97.54±00.24	98.17±00.19	90.57±00.35	91.02±00.48	87.98±00.42
Large Patch	94.17±00.35	94.53±00.26	93.71±00.31	98.74±00.15	97.98±00.23	98.39±00.18	90.12±00.41	91.02±00.52	87.69±00.49
Dynamic-scale	**94.60±00.33**	**94.88±00.29**	**94.14±00.36**	**99.02±00.12**	**98.24±00.33**	**98.69±00.16**	**90.95±00.30**	**91.43±00.70**	**88.11±00.37**

2) Dynamic-Scale Ablation Experiments: To verify the effectiveness of DSSM, we use fixed small-scale patches and fixed large-scale patches as comparisons to conduct ablation experiments. The dynamic scale for the Houston2013 dataset is the combination of 11 × 11 and 13 × 13, with fixed small-scale patch size referring to 11 × 11 and fixed large-scale patch size referring to 13 × 13. The dynamic scales for the Trento and MUUFL datasets are selected as a combination of 9 × 9 and 11 × 11, with fixed small-scale patch size referring to 9 × 9 and fixed large-scale patch size referring to 11 × 11. Table V shows that dynamic-scale patches perform best, proving that the DSSM can improve classification performance by adaptively managing the various scales of the objects.

D. Comparison Experiments

To verify the advancement and effectiveness of the proposed MADNet, we compare it with several state-of-the-art HSI and LiDAR fusion classification models, including the TBCNN [21], EndNet [63], multimodal deep learning framework (MDL-RS) [20], multiattentive hierarchical fusion net (MAHiDFNet) [64], dynamic-scale hierarchical fusion network (DSHFNet) [65], GLT-Net [29], and the MFT [25]. In order to ensure the fairness of the experiments, all the comparisons adopt the optimal parameters mentioned in the literature. We adopt the same training and test samples, with experiments in each network running 500 epochs and ten independent executions. The mean results with the standard deviation are presented, which can reduce the error caused by individual experimental changes and enhance the comparability between the methods.

1) Comparison of Different Classification Methods: The experimental results include both quantitative analysis and visualizations. The quantitative analysis results are presented from Tables VI to VIII, with the best accuracy for each row highlighted in bold. The visual results are depicted from Figs. 8 to 10, displaying the pseudo-color images for the HSI, LiDAR images, ground-truth images, and classification maps obtained by different methods. The colors used to represent each class are also indicated in the figures.

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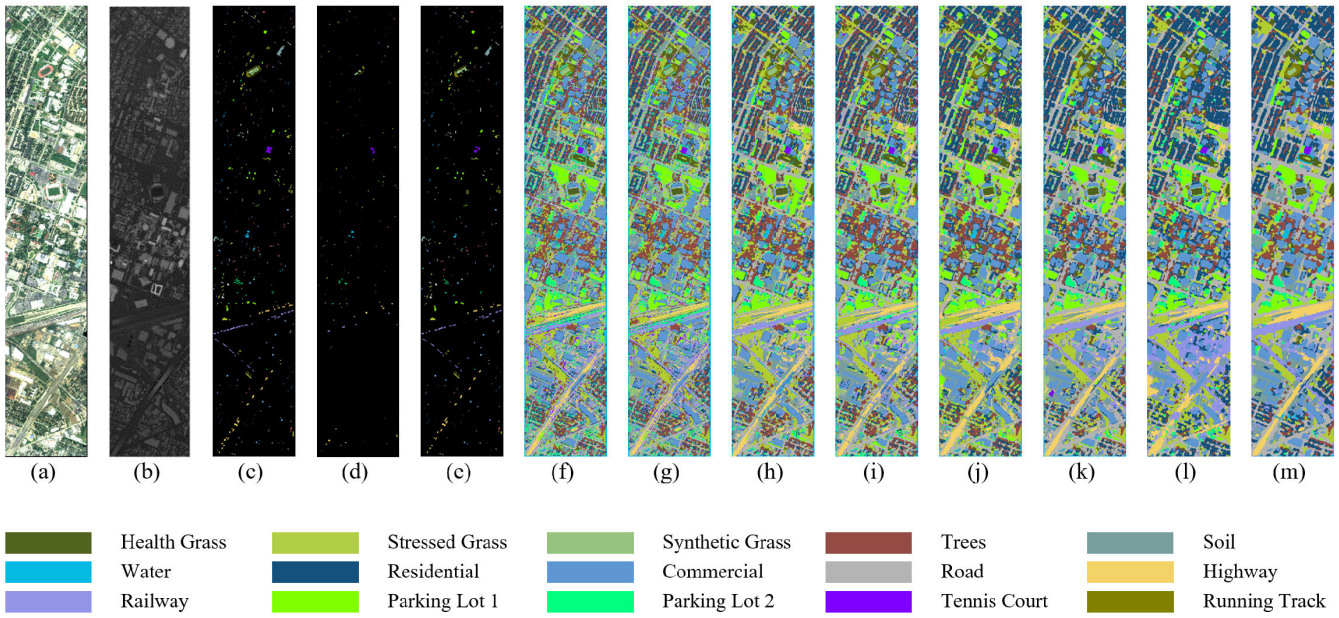


Fig. 8. Classification maps for the Houston2013 dataset obtained with different methods. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground truth. (d) Training samples. (e) Test samples. (f) TBCNN (86.61%). (g) EndNet (87.11%). (h) MDL-RS (90.80%). (i) MAHiDFNet (91.67%). (j) DSHFNet (92.87%). (k) GLT-Net (92.58%). (l) MFT (90.42%). (m) MADNet (94.60%).

TABLE VI
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Class No.	TBCNN [21]	EndNet [63]	MDL-RS [20]	MAHiDFNet [64]	DSHFNet [65]	GLT-Net [29]	MFT [25]	MADNet
1	84.43±00.74	83.34±00.66	**85.31±02.58**	82.99±01.95	83.57±00.95	83.41±02.47	83.07±02.27	81.51±00.44
2	97.87±01.25	98.81±00.90	98.04±01.64	97.64±02.22	99.45±00.63	98.53±03.90	97.52±02.87	**99.95±00.08**
3	99.52±00.39	99.37±00.50	**99.86±00.20**	94.18±01.90	95.45±02.20	97.64±01.20	98.85±01.47	93.05±01.22
4	99.43±00.37	99.53±00.31	99.57±00.60	99.14±00.62	**99.87±00.15**	99.83±00.21	97.92±01.11	97.54±00.83
5	96.95±03.64	96.66±02.34	99.86±00.22	98.98±00.64	**99.98±00.06**	99.66±00.54	99.50±00.50	99.71±00.21
6	92.66±03.87	90.98±02.29	95.17±02.24	92.38±03.73	96.15±01.86	97.76±02.00	96.50±02.17	**98.25±01.66**
7	87.41±01.54	86.61±01.31	91.53±05.30	93.83±02.05	**94.89±01.33**	93.03±02.43	93.22±01.73	93.81±01.60
8	87.53±06.59	87.83±05.52	84.77±07.42	89.72±03.72	87.11±02.09	90.09±03.90	89.32±04.11	**92.68±01.81**
9	79.58±07.73	79.81±06.18	74.68±12.72	88.46±03.66	87.73±01.70	88.85±02.68	88.11±03.03	**96.34±01.40**
10	60.25±12.07	66.32±10.31	88.11±07.28	86.81±06.55	87.94±02.40	86.94±05.19	66.29±11.41	**92.71±02.93**
11	92.10±02.83	92.86±02.20	89.49±08.26	94.90±02.80	95.63±01.29	94.43±02.35	96.16±01.79	**98.89±00.78**
12	62.68±18.06	62.30±17.63	86.66±04.56	84.03±03.01	88.53±02.49	85.04±03.00	88.44±04.51	**91.95±02.42**
13	**91.58±00.95**	84.73±03.30	89.82±02.36	90.28±01.58	88.21±03.79	89.61±03.26	82.81±05.05	89.75±04.17
14	99.64±00.12	99.55±00.22	97.77±02.46	91.66±04.39	96.88±02.36	96.68±03.45	96.88±07.91	**99.96±00.12**
15	99.20±00.76	98.54±01.04	**99.64±00.47**	92.60±05.45	97.12±02.82	97.95±03.65	98.75±02.04	97.12±02.73
OA (%)	86.61±00.78	87.11±01.26	90.80±01.72	91.67±00.51	92.87±00.42	92.58±00.68	90.42±00.98	**94.60±00.33**
AA (%)	88.72±00.63	88.92±01.01	91.91±01.37	91.47±00.39	93.34±00.52	93.30±00.72	91.29±01.06	**94.88±00.29**
Kappa×100	85.47±00.85	86.01±01.37	90.01±01.87	90.95±00.55	92.26±00.46	91.94±00.74	89.60±01.06	**94.14±00.36**

The Houston2013 dataset consists of 15 classes and exhibits complex scenes. The classification difficulty of highways and surrounding objects is increased due to the influence of cloud shadows. From Table VI, we can draw the following conclusions. First, by comparing several networks based on CNN, we find that using attention mechanism can get more accurate classification results. For example, both MAHiDFNet and DSHFNet are CNN networks with the attention mechanism, while MDL-RS is a pure CNN network. MAHiDFNet and DSHFNet have higher classification accuracy than MDL-RS, indicating that attention mechanisms can make networks pay more attention to useful information. Second, multiscale feature extraction is more effective than single-scale feature extraction. For example, DSHFNet and GLT-Net are multiscale networks, and their classification performance on the Houston2013 dataset is higher than that of other single-scale networks, indicating that multiscale networks can take full advantage of information on different scales to get superior classification results. Finally, if a combined CNN and Transformer network is used, the

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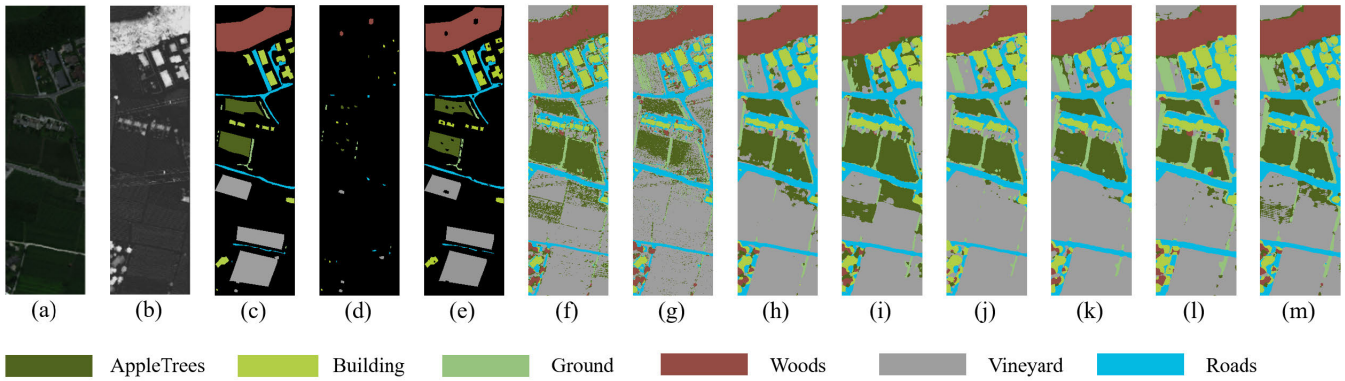


Fig. 9. Classification maps for the Trento dataset obtained with different methods. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground truth. (d) Training samples. (e) Test samples. (f) TBCNN (96.62%). (g) EndNet (95.80%). (h) MDL-RS (98.17%). (i) MAHiDFNet (98.14%). (j) DSHFNet (98.84%). (k) GLT-Net (98.75%). (l) MFT (97.31%). (m) MADNet (99.02%).

TABLE VII
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE TRENTO DATASET

Class No.	TBCNN [21]	EndNet [63]	MDL-RS [20]	MAHiDFNet [64]	DSHFNet [65]	GLT-Net [29]	MFT [25]	MADNet
1	96.45±01.04	93.96±02.57	98.09±04.51	99.63±00.18	**99.90±00.11**	98.75±00.33	95.69±01.02	99.02±00.33
2	97.73±00.68	97.12±00.75	98.24±01.69	99.66±00.23	99.66±00.12	99.67±00.30	**99.87±00.14**	99.79±00.14
3	95.29±02.49	84.09±03.76	90.27±15.46	72.81±05.14	84.57±05.01	92.70±04.73	88.07±09.45	**98.34±01.48**
4	99.67±00.08	99.74±00.11	99.62±00.54	99.85±00.20	**100.00±00.00**	**100.00±00.00**	99.98±00.04	**100.00±00.00**
5	95.61±00.34	93.52±01.28	99.10±01.67	99.91±00.08	99.97±00.03	99.63±00.56	99.86±00.12	**99.99±00.02**
6	90.44±01.18	**94.52±01.06**	91.74±04.16	86.90±02.62	91.22±01.49	92.05±02.05	81.74±03.21	92.27±01.25
OA (%)	96.62±00.18	95.80±00.43	98.17±00.99	98.14±00.30	98.84±00.14	98.75±00.29	97.31±00.31	**99.02±00.12**
AA (%)	95.87±00.47	93.82±00.74	96.18±03.02	93.13±00.90	95.89±00.76	97.13±00.82	94.20±01.76	**98.24±00.33**
Kappa×100	95.49±00.24	94.39±00.58	97.55±01.33	97.50±00.40	98.44±00.19	98.33±00.39	96.40±00.42	**98.69±00.16**

classification results will be better than if only CNN or Transformer is used alone. For example, GLT-Net combines the advantages of CNN and Transformer, achieving better performance than the merely CNN-based or Transformer-based networks.

Particularly, even with the complexity of the Houston2013 dataset, the MADNet achieves an OA of 94.60%, AA of 94.88%, and Kappa of 94.14%, which are 1.73%, 1.54%, and 1.88% higher than the suboptimal approach, respectively. The classification results for categories such as Stressed grass, Water, Commercial, Road, and Highway are superior to other methods. It is worth noting that, due to the influence of cloud shadow, Highway and Railway appear the phenomenon of the same object with abnormal spectrum, resulting in poor performance in most methods. Benefiting from the SAAM, our network can well distinguish objects disturbed by cloud shadows. In addition, it can be found that the classification results of MADNet are acceptable for each class, which means that our proposed DSSM can effectively adapt to the scale characteristics of different land cover categories. Furthermore, from the visualizations in Fig. 8, we can see that mutation categories rarely appear in the middle of each continuous category, because the DWBT reduces the redundancies of the remote information.

For the Trento dataset, as shown in Table VII, the MADNet achieves an OA of 99.02%, AA of 98.24%, and Kappa of 98.69%, which are 0.18%, 1.11%, and 0.36% higher than the suboptimal results, respectively. Our model reaches the optimal level for half of the six classes in this dataset. As can be seen from Table VII, the accuracy of the Ground category is significantly improved. The visual result also shows that the boundaries of the Ground around the Apple trees are much clearer, demonstrating the superiority of our method.

Table VIII presents the classification accuracies of all comparisons on the MUUFL dataset. The MADNet achieves the highest OA, AA, and Kappa of 90.95%, 91.43%, and 88.11%, respectively, compared to the other competing methods. From the visual results, we can see that the classification map produced by our method is apparently smooth, and there are few isolated classification points, which implies that the DWBT can effectively reduce remote redundant information. In summary, the experiments on all three datasets consistently demonstrate the advancement of our approach.

2) Discussion on Different Sampling Strategies: Some works [66], [67] point out that the random sampling strategy has a great influence on the reliability and quality of the obtained solution, because it inevitably suffers from pixel-sharing problems between neighboring training and test samples to a certain extent. In this section, we resample the MUUFL dataset with a controlled random sampling strategy (CRSS) [68] to demonstrate differences from previous random sampling experiments. 5% labeled samples are selected to train the networks. The results are shown in Fig. 11 and Table IX.

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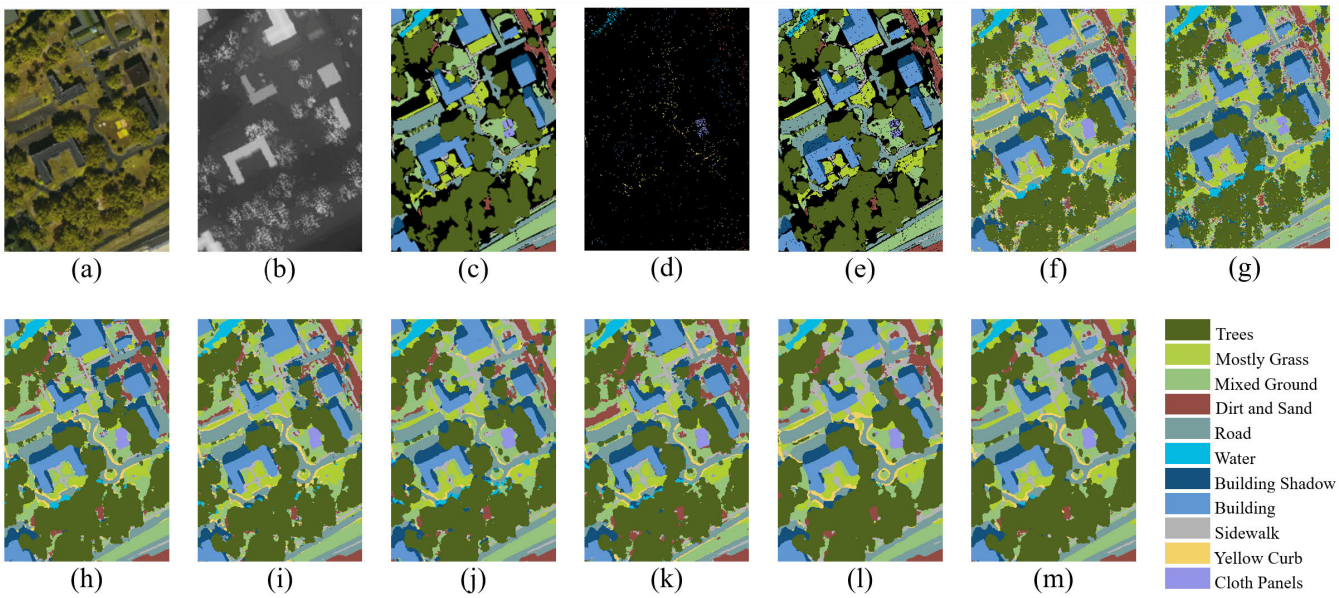


Fig. 10. Classification maps for the MUUFL dataset obtained with random sampling strategy. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground truth. (d) Train samples. (e) Test samples. (f) TBCNN (85.67%). (g) EndNet (81.63%). (h) MDL-RS (88.34%). (i) MAHiDFNet (88.66%). (j) DSHFNet (89.42%). (k) GLT-Net (90.80%). (l) MFT (88.06%). (m) MADNet (90.95%).

TABLE VIII
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE MUUFL DATASET WITH RANDOM SAMPLING STRATEGY

Class No.	TBCNN [21]	EndNet [63]	MDL-RS [20]	MAHiDFNet [64]	DSHFNet [65]	GLT-Net [29]	MFT [25]	MADNet
1	88.66±01.70	83.51±01.42	91.17±02.29	93.08±01.79	93.28±00.96	**94.21±00.94**	91.71±01.16	93.97±01.12
2	80.64±05.51	82.94±02.76	**85.45±04.50**	81.06±03.53	78.95±06.42	80.85±04.55	78.88±04.14	83.31±04.86
3	68.31±05.70	59.73±02.41	72.29±06.95	72.64±04.92	76.52±03.77	79.43±04.63	77.74±03.39	**80.34±05.74**
4	84.17±07.12	82.64±03.52	**97.34±01.57**	90.68±07.12	95.43±02.16	95.86±02.46	92.56±01.84	94.85±02.36
5	87.87±01.78	88.05±01.10	88.28±03.03	88.39±03.56	88.69±01.23	90.38±01.48	86.57±02.20	**90.39±01.89**
6	99.48±00.41	96.94±01.11	**100.00±00.00**	98.83±03.26	99.97±00.08	99.73±00.66	99.75±00.31	99.92±00.13
7	92.30±01.40	85.19±01.68	**96.01±04.84**	91.05±04.35	92.81±01.22	93.70±02.57	90.90±02.04	94.06±01.91
8	93.80±01.32	90.39±01.12	92.67±01.88	94.90±00.91	95.67±00.83	95.30±01.23	94.33±00.78	**95.96±01.09**
9	75.04±09.33	70.07±03.21	70.87±08.37	79.13±04.68	77.65±03.29	**84.43±03.55**	68.69±04.80	82.21±03.66
10	91.44±04.68	92.17±03.01	93.01±02.73	88.56±02.71	92.53±02.79	**95.66±01.96**	90.36±02.16	92.29±04.76
11	93.39±01.64	95.15±01.49	**99.29±00.36**	98.46±00.85	98.70±00.52	98.52±01.46	98.76±01.31	98.46±00.80
OA (%)	85.67±00.73	81.63±00.77	88.34±01.45	88.66±00.45	89.42±00.37	90.80±00.42	88.06±00.45	**90.95±00.30**
AA (%)	87.10±00.92	84.25±00.52	89.85±00.84	88.80±01.63	90.02±00.55	91.39±00.47	88.21±00.55	**91.43±00.70**
Kappa×100	81.39±00.87	76.43±00.92	84.81±01.79	85.12±00.56	86.11±00.45	87.90±00.53	84.40±00.56	**88.11±00.37**

It can be concluded that our network still performs best even without the influence of overlapping training and test samples, especially in the classification of dirt and sand. It means that our method can still distinguish each class by spectral features in the case of loss of spatial information.

3) Performance With Different Training Sample Volumes: To further validate the robustness and effectiveness of the proposed method, we investigated the classification accuracy of the model when the sample size is smaller. Specifically, we randomly select a fixed percentage of samples from each category in the training set for training, while keeping the test set unchanged. The training set percentage is increased from 20% to 100%. Fig. 12 shows the varying classification performance of the proposed MADNet model and other comparison methods, described by the average OA over ten independent repeated experiments. From Fig. 12, it can be seen that as the number of training samples reduces, the classification accuracies of all methods decrease, indicating that the models rely on sufficient sample sizes for training. It is worth noting that our model always achieves optimal performance when the number of training samples is the same, which proves the robustness of the frameworks.

4) Computational Complexity Analysis: In order to comprehensively measure the performance of the model, the complexity should also be considered. Specifically, the number of trainable parameters and computational cost are effective indexes for measuring the spatial and temporal complexity of the network, respectively. Fig. 13(a)–(c) illustrates the OA, parameters, and computational costs of the competitive networks on the three datasets. The horizontal coordinate

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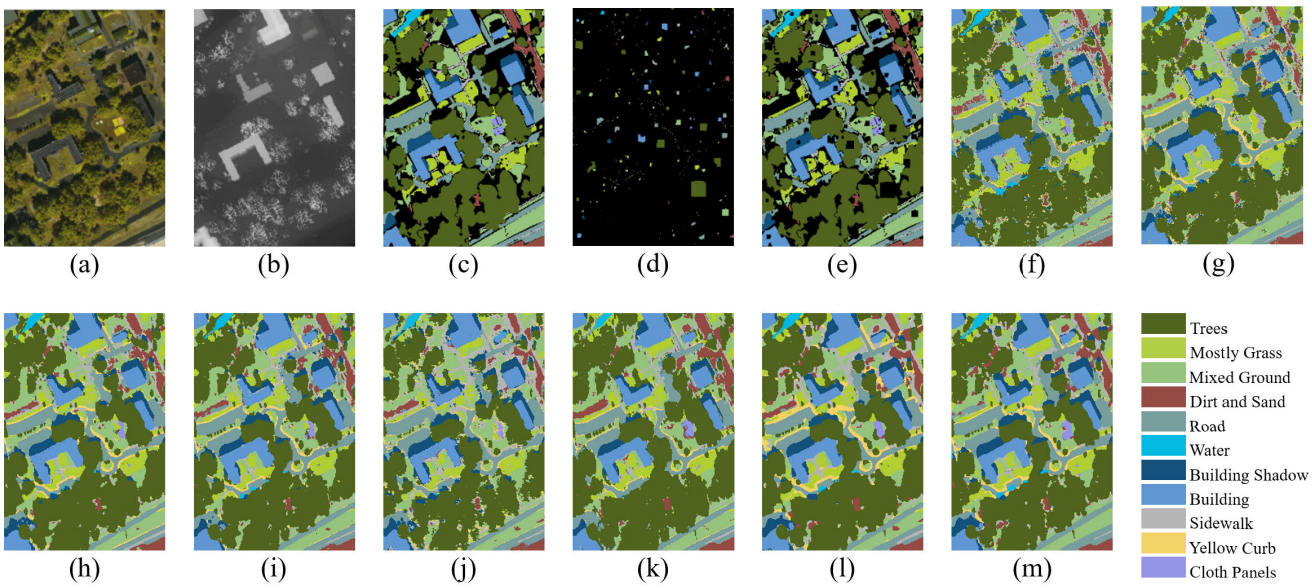


Fig. 11. Classification maps for the MUUFL dataset obtained with controlled random sampling strategy. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground truth. (d) Training samples. (e) Test samples. (f) TBCNN (87.72%). (g) EndNet (84.52%). (h) MDL-RS (87.39%). (i) MAHiDFNet (87.67%). (j) DSHFNet (87.71%). (k) GLT-Net (87.77%). (l) MFT (85.76%). (m) MADNet (88.44%).

TABLE IX
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE MUUFL DATASET WITH CONTROLLED RANDOM SAMPLING STRATEGY

Class No.	TBCNN [21]	EndNet [63]	MDL-RS [20]	MAHiDFNet [64]	DSHFNet [65]	GLT-Net [29]	MFT [25]	MADNet
1	94.82±00.65	94.48±00.63	**96.05±00.33**	95.92±00.24	94.19±00.64	93.84±00.75	93.82±00.78	95.12±00.70
2	77.70±03.73	63.88±03.72	74.43±02.47	72.91±04.08	77.37±04.72	64.94±06.57	72.32±05.41	**79.39±05.19**
3	75.28±02.73	74.23±01.89	78.32±01.91	78.65±01.77	75.80±04.36	**91.19±00.54**	82.36±05.36	81.90±03.14
4	81.93±04.68	68.66±05.02	77.66±03.55	78.98±01.91	82.60±02.57	77.58±02.81	81.44±07.14	**85.13±03.45**
5	88.18±01.45	82.72±01.10	82.88±01.22	84.17±00.88	**88.87±01.31**	82.26±02.49	84.57±03.23	83.72±02.93
6	69.80±03.48	56.93±05.91	65.53±03.67	60.86±02.90	71.65±03.11	71.56±02.85	58.46±10.18	**74.49±02.21**
7	78.27±02.67	72.28±02.77	74.49±01.62	75.61±02.01	78.92±01.34	79.76±03.15	77.59±04.34	**84.66±03.76**
8	**94.65±00.39**	93.06±01.03	93.12±00.59	93.28±01.02	94.52±00.67	93.88±01.30	91.55±01.82	90.29±01.18
9	66.04±05.90	62.85±02.25	68.80±03.10	**77.14±00.84**	68.68±02.73	70.82±09.99	63.71±06.83	68.91±09.66
10	85.84±02.51	89.60±03.61	92.88±01.62	96.88±00.75	85.28±01.61	**97.60±00.56**	67.20±16.90	94.27±02.23
11	38.86±14.48	26.26±03.79	59.17±08.22	51.02±10.79	43.74±10.46	**78.35±02.41**	63.39±16.03	71.19±02.95
OA (%)	87.72±00.27	84.52±00.42	87.39±00.35	87.67±00.30	87.71±00.28	87.77±00.63	85.76±00.65	**88.44±00.28**
AA (%)	77.40±01.87	71.36±01.10	78.48±01.09	78.67±01.08	78.33±00.97	80.98±02.18	75.58±02.68	**82.64±01.34**
Kappa×100	83.80±00.35	79.62±00.55	83.37±00.46	83.72±00.41	83.84±00.35	83.99±00.85	81.31±00.91	**84.83±00.35**

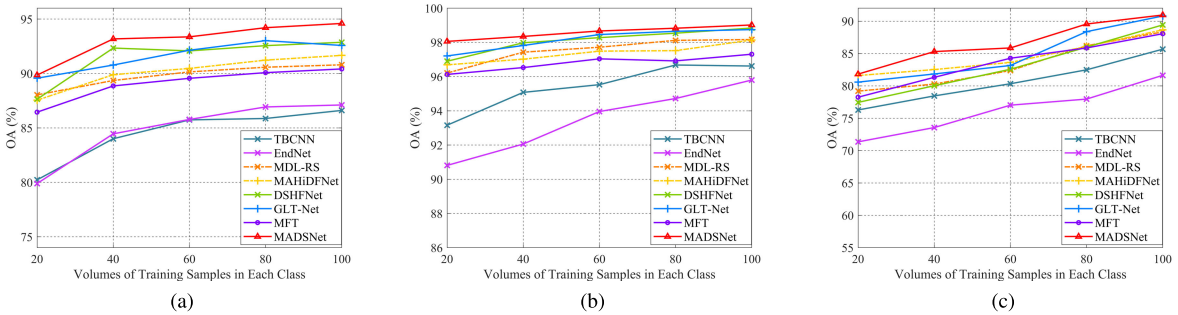


Fig. 12. Classification performance with different volumes of training samples. (a) Houston2013, (b) Trento, and (c) MUUFL.

represents the number of trainable parameters, the vertical coordinate represents the OA, and the bubble diameter represents the computational cost by million floating-point operations (MFLOPS). It is observed that the MDL-RS has fewer parameters and lower computational cost, but the OA is comparatively inferior. The MFT has more parameters and

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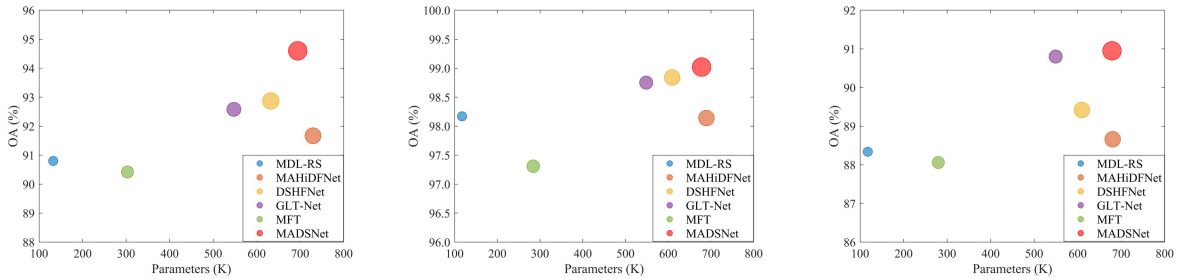


Fig. 13. Comparison of the networks on trainable parameters, computational cost, and OA. (a) Houston2013, (b) Trento, and (c) MUUFL.

higher computational cost than the MDL-RS, but the OA is lower, indicating that its network design is not that good. The MAHiDFNet has an improvement over the MDL-RS in OA, but the cost significantly increases the model complexity. The GLT-Net, DSHFNet, and MADNet have higher complexity but the OAs are higher than the MDL-RS. Although the proposed MADNet occupies the most computational cost, its number of parameters is less than the MAHiDFNet, and it achieves the best OAs on all three datasets.

V. Conclusion

In this article, we propose a novel MADNet to improve the performance of land use/land cover classification with the HSI and LiDAR data. To make full use of features with different depths, we design a multilevel attention mechanism. For the low-level natural spectral features, the SAAM can highlight the neighbor pixels that are spectrally similar to the central pixel. For the high-level multimodal fusion features, the GLCAM can learn the relationships between the pixels of patches. In particular, the deployment of DWBT in GLCAM can reduce remote redundant information and further improve the classification performance. Aiming at the problem that various categories of objects have different scales in remote sensing classification tasks, the DSSM is designed to dynamically select the patches of appropriate scales by analyzing pixel similarity statistics. The compared experiments and ablation studies on three benchmark HSI-LiDAR datasets consistently demonstrate the superiority of the proposed frameworks.

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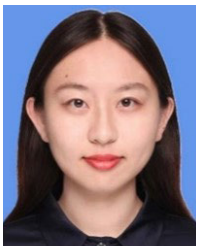
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